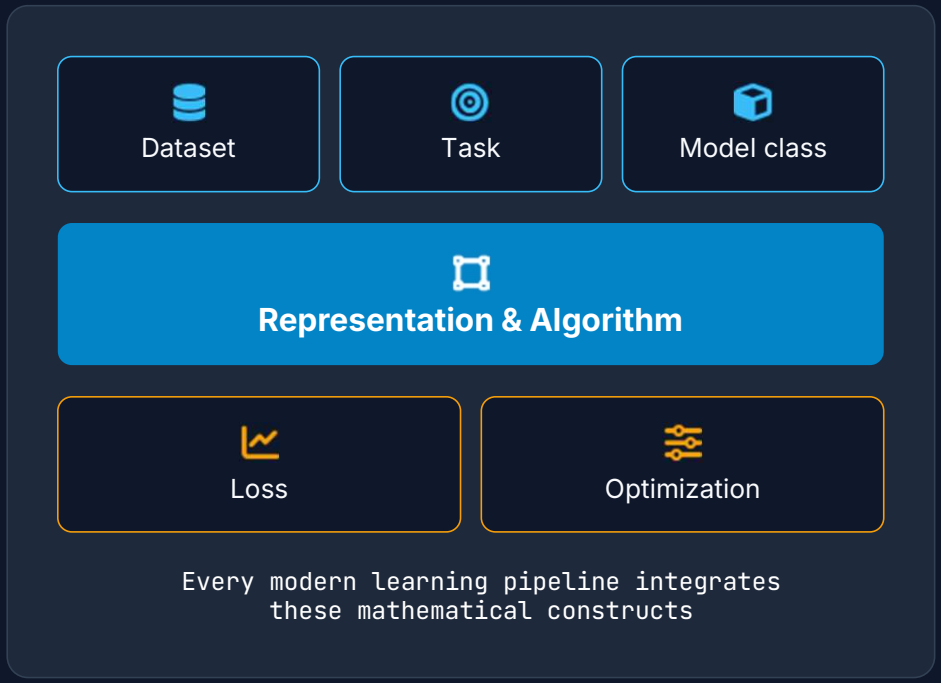
Session 03 / Components, Tasks, and Models

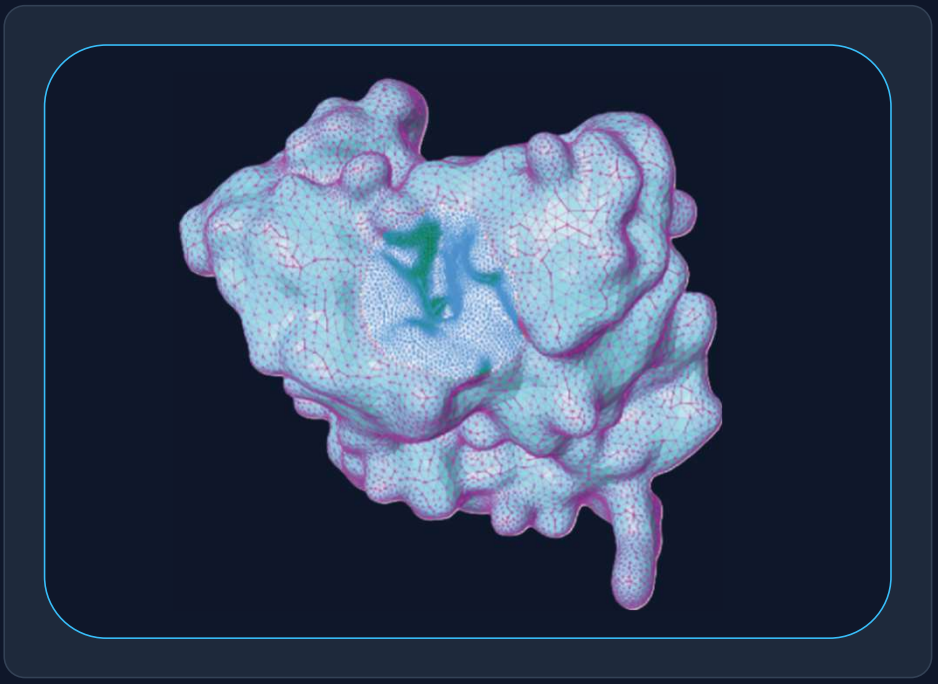
The ML Landscape

Mohammad Mahdi Nasimi

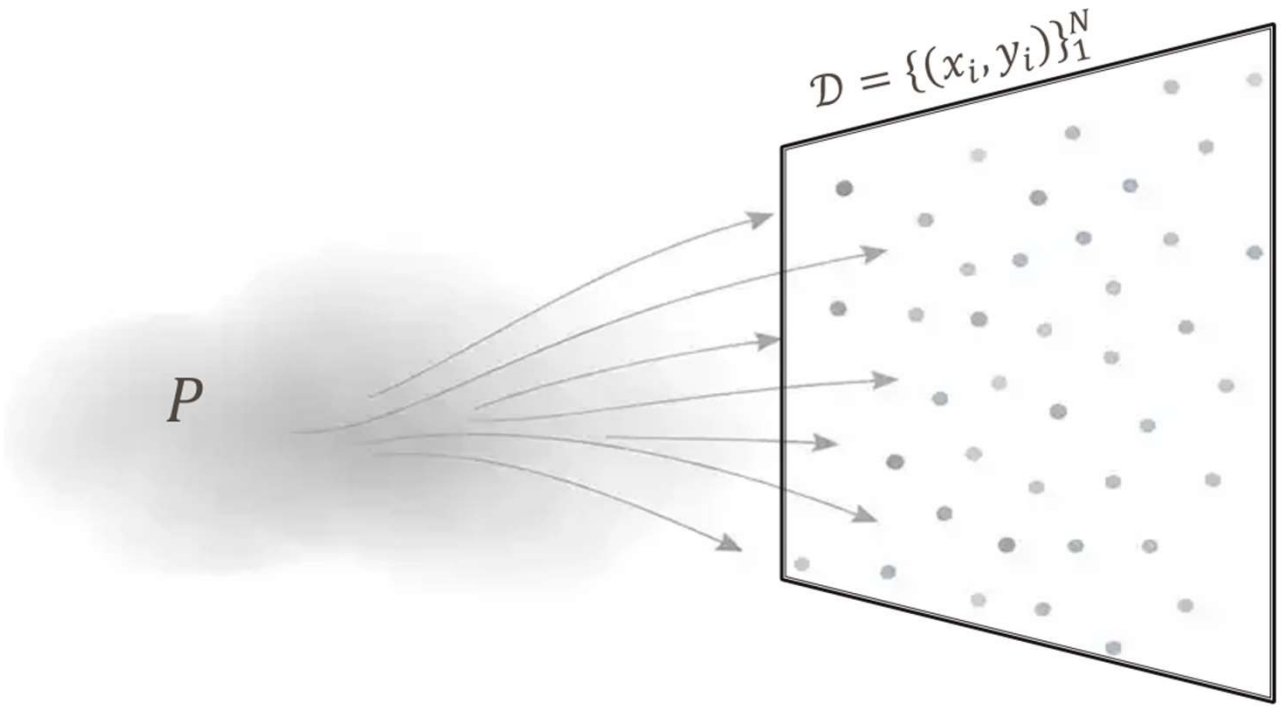


PART 1

A Reminder: ML is a Representation Problem



The physical world and raw data

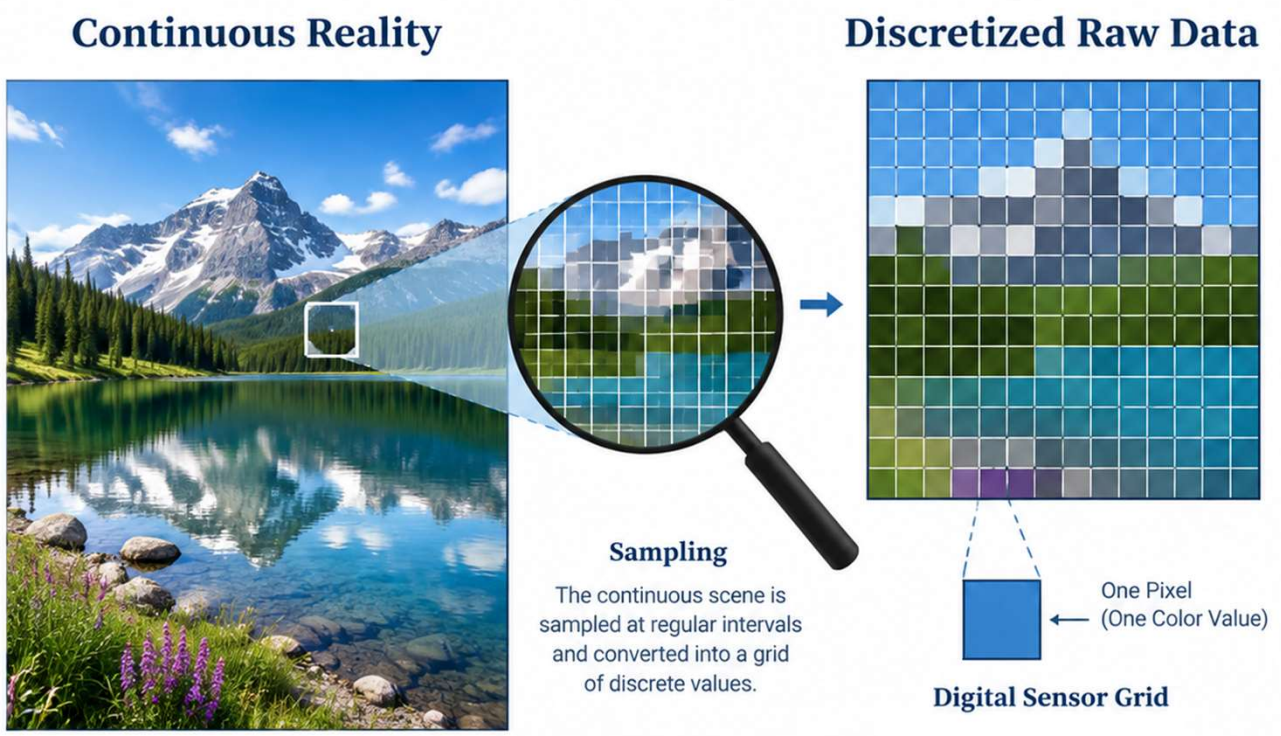


Finite observations

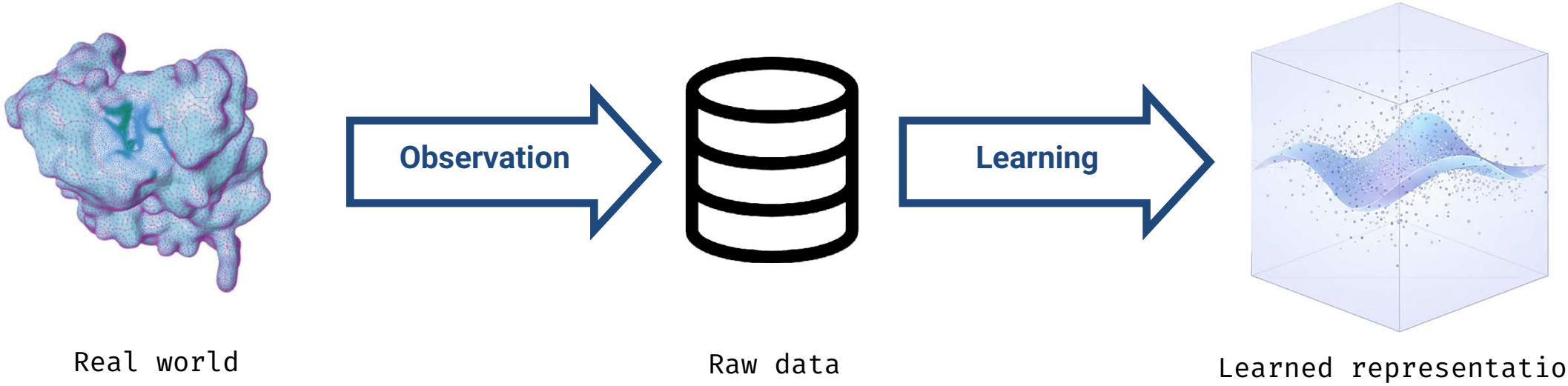
Sensors sample a richer phenomenon.

Precision, hidden variables and noise limit what we observe.

Sampling a continuous scene



Observation and representation



Two maps. Two chances to lose what matters.

The representation problem

The representation problem. Given a phenomenon P with structure $\mathcal{S}$ and a task T, an observation map $o: P \to \mathbb{R}^n$, find a representation $\phi: \mathbb{R}^n \to \mathbb{R}^m$ such that the structure relevant to T is preserved or made explicit in the coordinates $\phi(o(P))$.

The representation ϕ is the object of study. The algorithm $\mathcal{A}$ is the procedure that finds ϕ.

The rest of the course is the systematic study of $\mathcal{A}$.

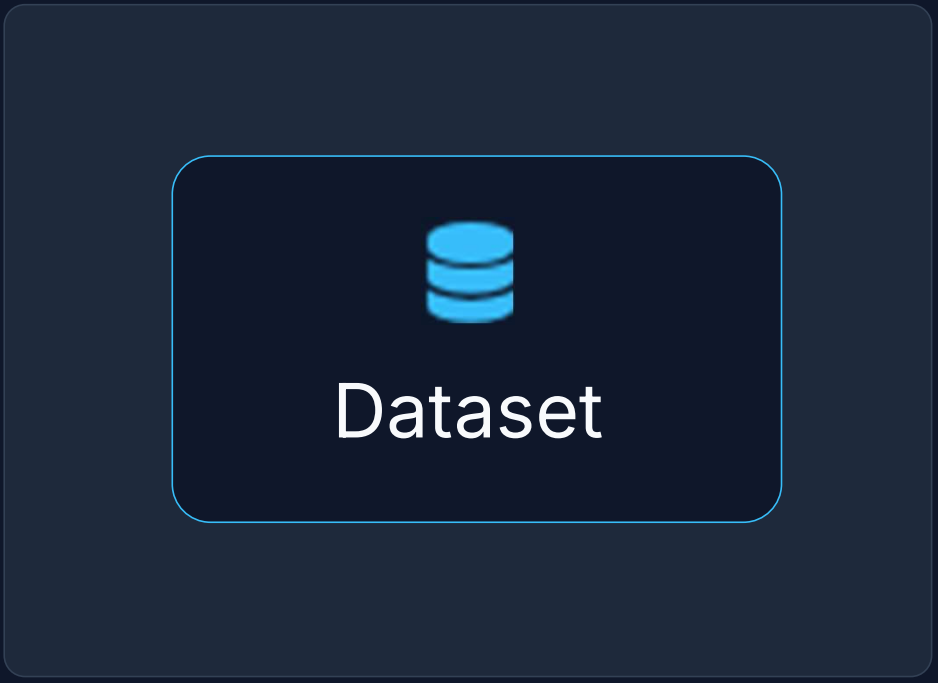
SLIDE 03 // PART 1 – MAIN COMPONENTS OF ML

Main Components of Machine Learning



PART 2

Main Components of ML: 1. Dataset



SLIDE 5 // PART 2 – ML DATASETS

Kinds of Datasets



Tabular



Image



Text



Time Series



Graph



Audio



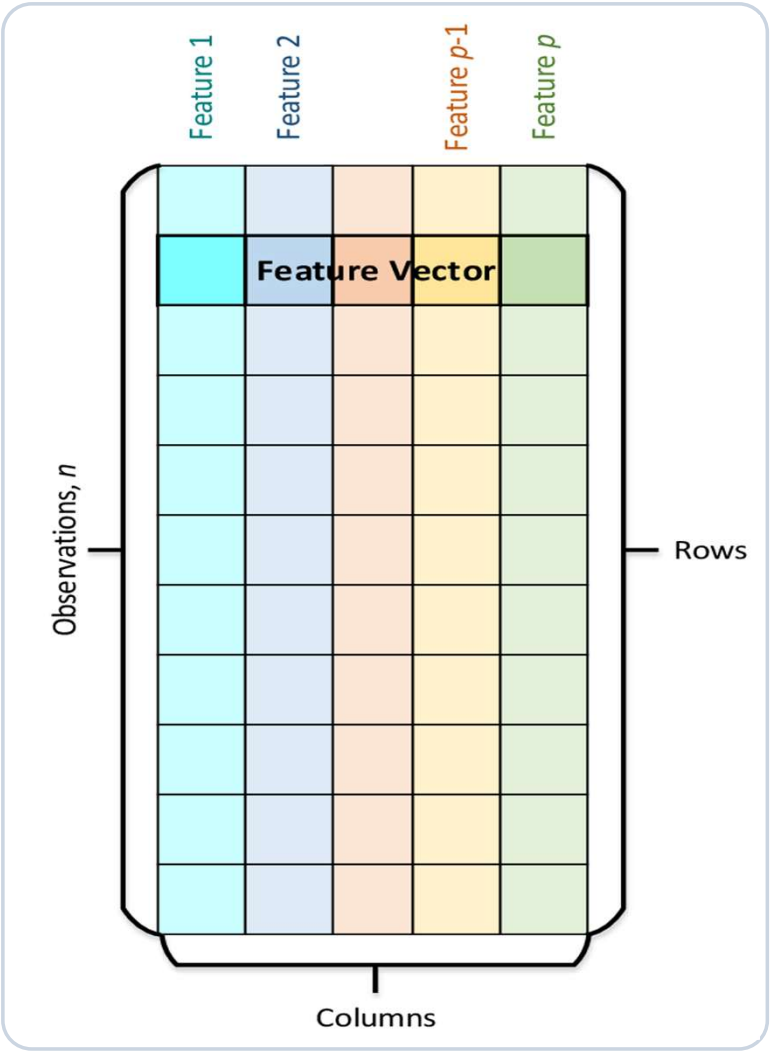
Video

SLIDE 6 // COMPONENT 01

Dataset

Matrix Structure

- Rows = observations,
- Columns = features



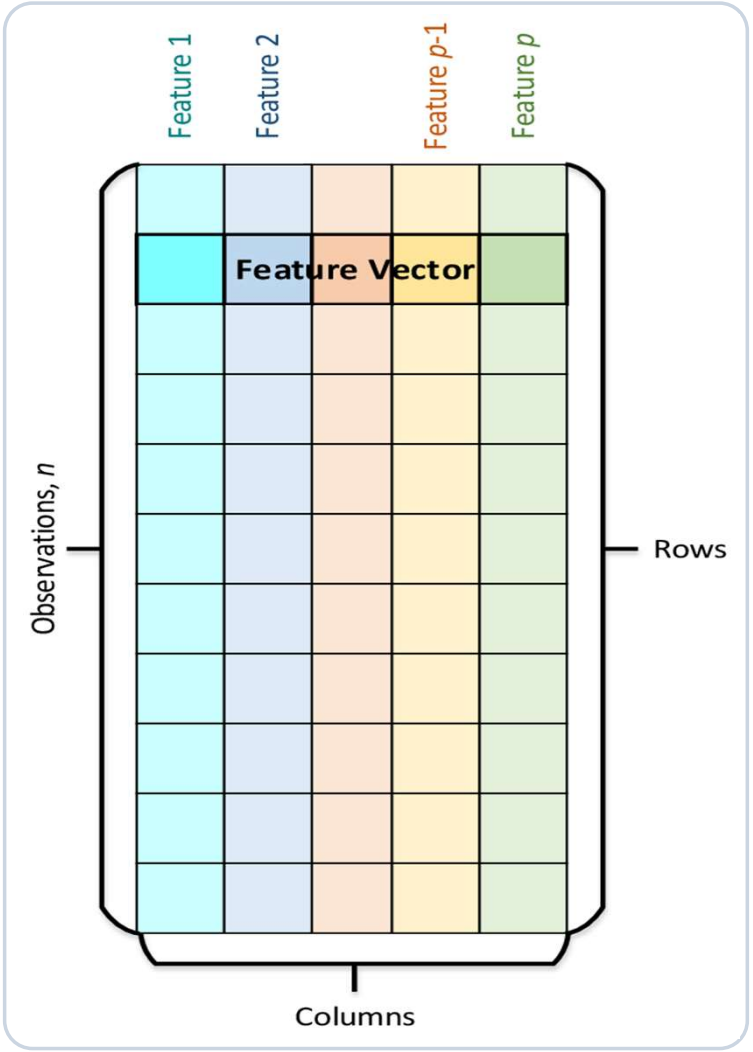
SLIDE 7 // COMPONENT 01

Dataset

Supervised Data: $\mathcal{D} = \{(x_i, y_i)\}_1^n$

Unsupervised Data: $\mathcal{D} = \{x_i\}_1^n$

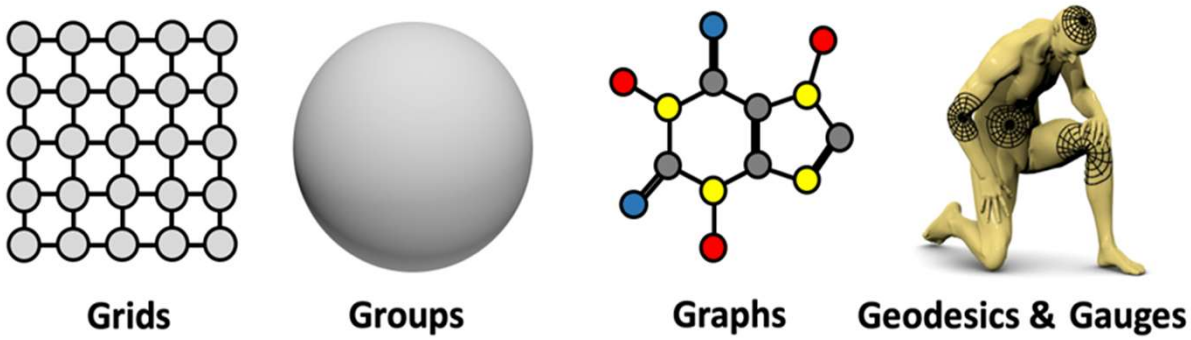
Design Matrix: $X \in \mathbb{R}^{N \times D}$



SLIDE 8 // COMPONENT 01

Dataset

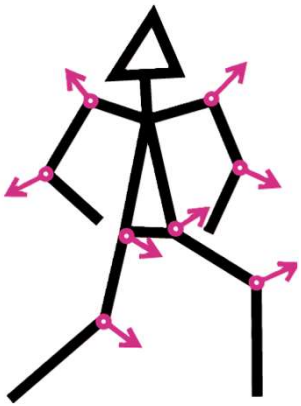
Ultimately, computer memory is just a table (or grid of bytes). So how can complex structures like a graph, a colored image, or a 3D mesh be stored in a table?

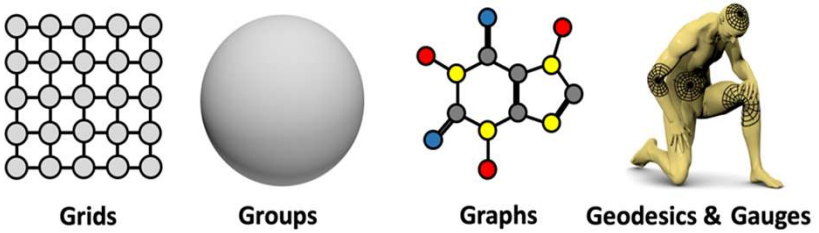


SLIDE 9 // COMPONENT 01

Dataset

Ultimately, computer memory is just a table (or grid of bytes). So how can complex structures like a graph, a colored image, or a 3D mesh be stored in a table?





SLIDE 10 // PART 2 – BENCHMARKS

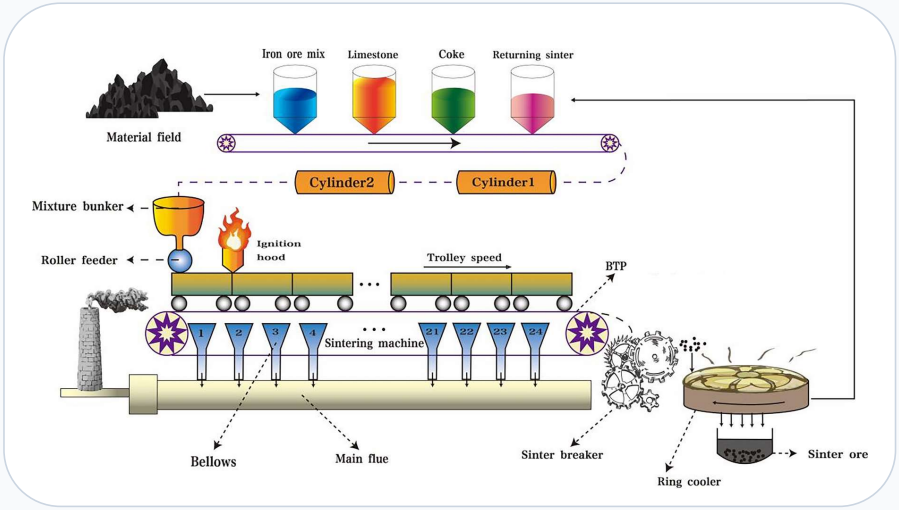
Dataset Examples

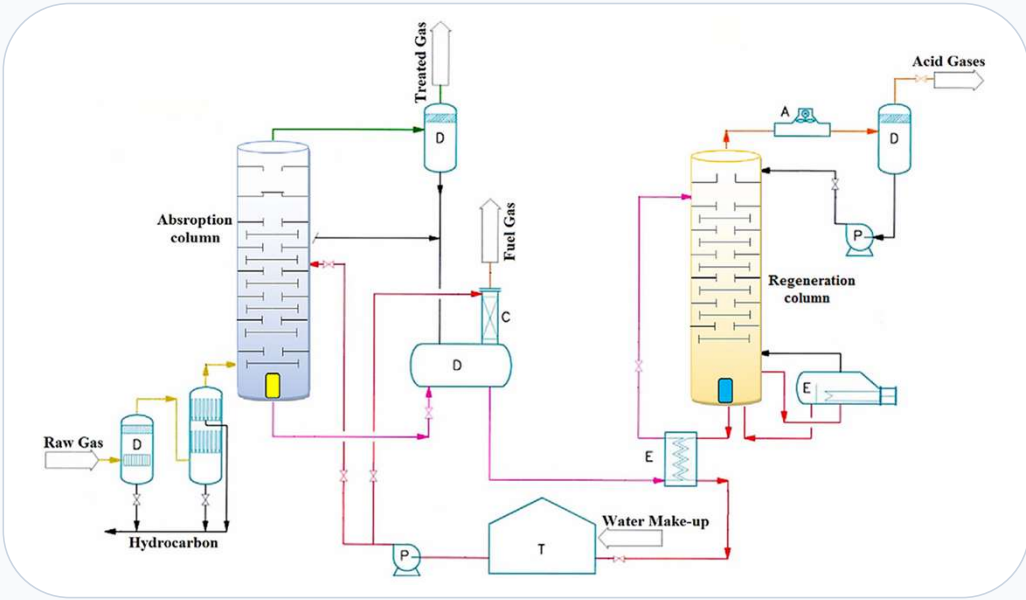
Category	Example 1	Example 2
Tabular	Iris	Boston Housing
Image	MNIST	CIFAR-10
Text	IMDB Reviews	Wikipedia
Time series	Stock prices	ECG signals
Graph	Social networks	Molecules
Audio	Speech commands	Music
Video	Action recognition	Autonomous driving

SLIDE 11 // COMPONENT 01

Dataset

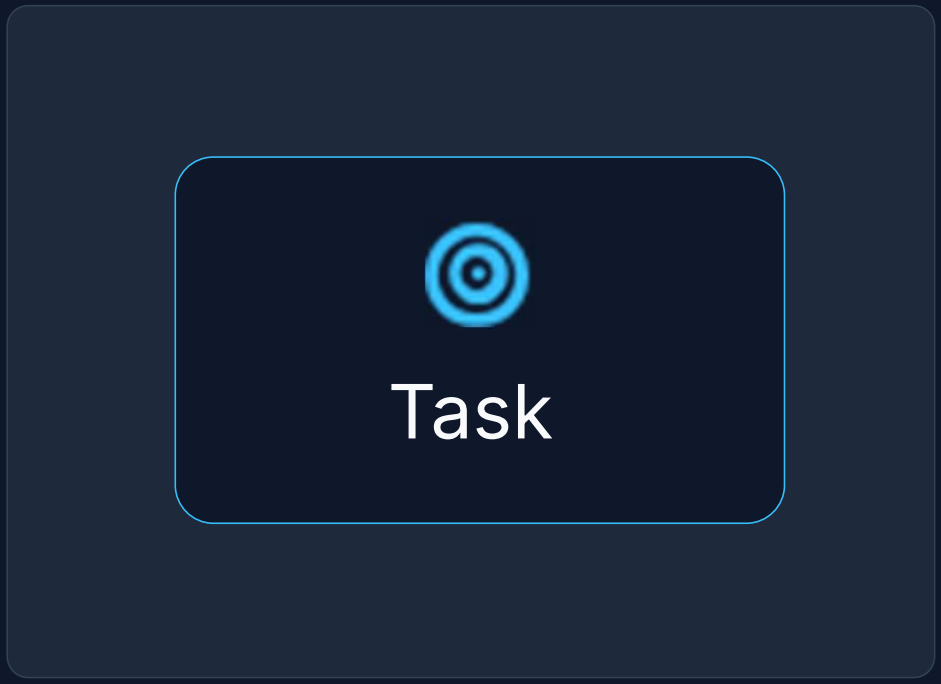
Let’s see some real datasets.





PART 3

Main Components of ML: 2. Task



SLIDE 13 // COMPONENT 02

Task definition

Task:

- What we want to predict or discover

+

- it’s evaluation metrics

Task examples

- **Real Estate:** Predict house prices
- **Security:** Classify emails as spam or not
- **Analytics:** Group customers by behavior
- **Grid Monitoring:** Detect anomalies in a power grid
- **Control:** Play Go
- **Language:** Translate French to English
- **Synthesis:** Generate images from text

Task examples

- **Real Estate:** Predict house prices
- **Security:** Classify emails as spam or not
- **Analytics:** Group customers by behavior
- **Grid Monitoring:** Detect anomalies in a power grid
- **Control:** Play Go
- **Language:** Translate French to English
- **Synthesis:** Generate images from text

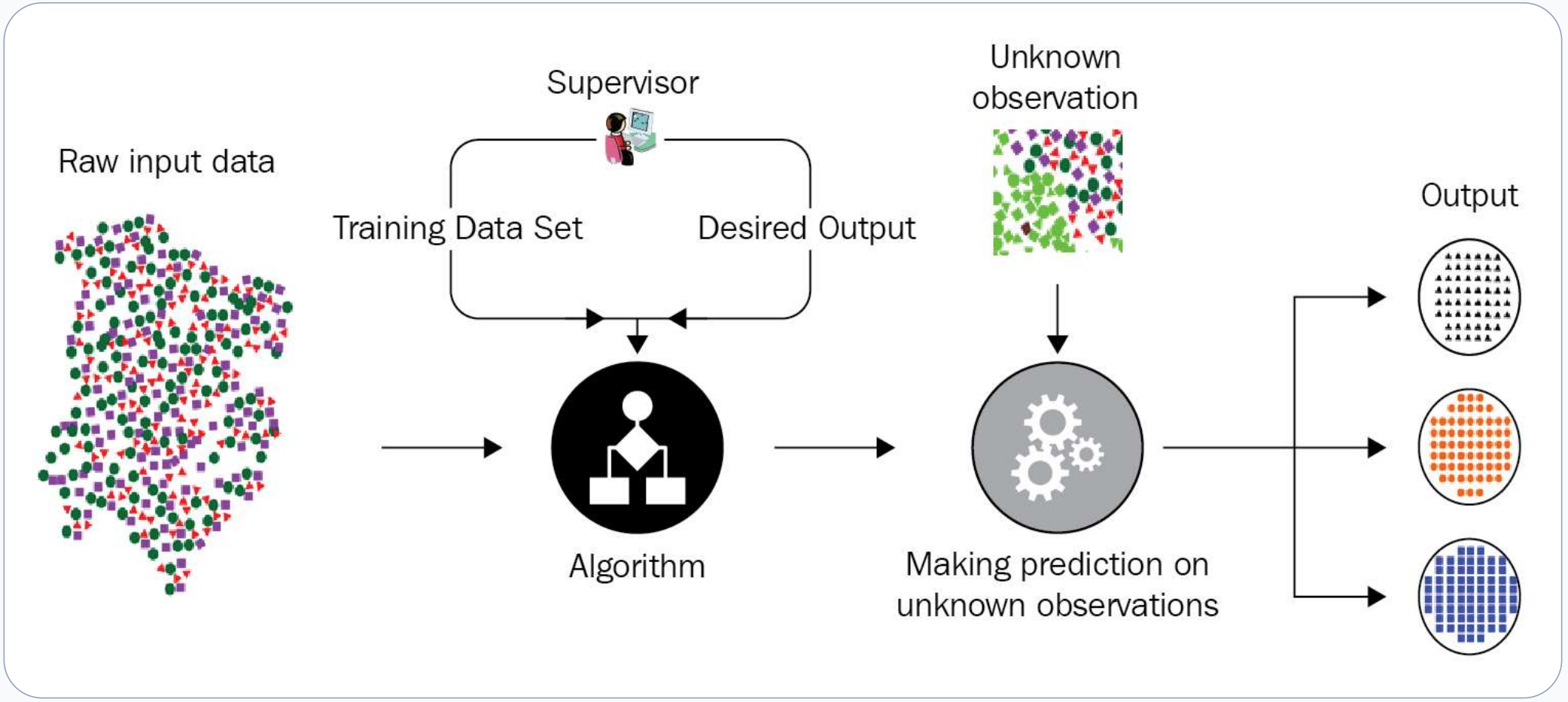
Group

- How would you cluster them into families?
- What distinguishes one group from another?

Discuss with your groupmate for 3 minutes

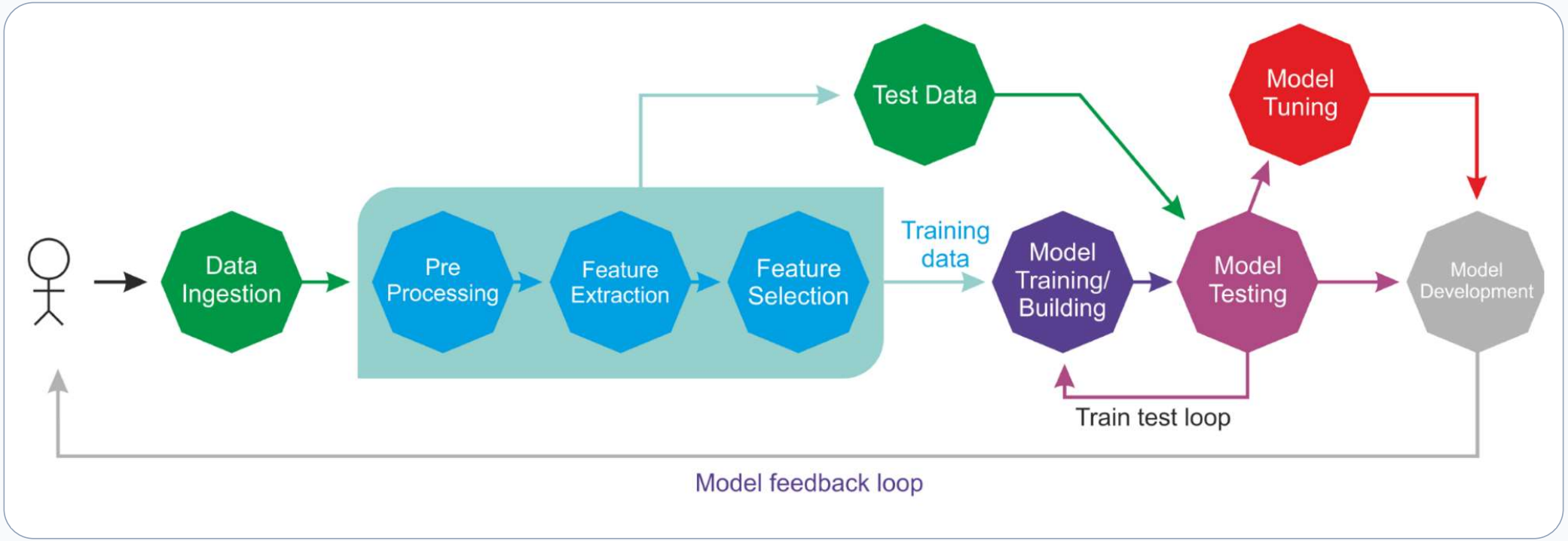
SLIDE 16 // TASK TAXONOMY

Supervised



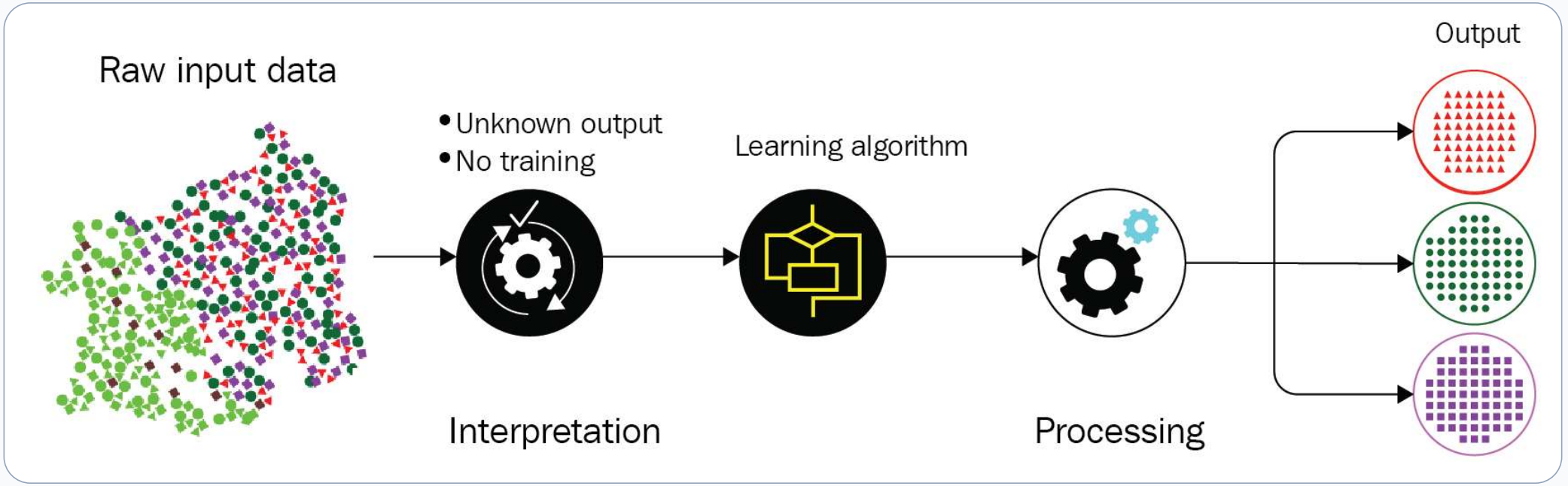
SLIDE 17 // TASK TAXONOMY

Supervised

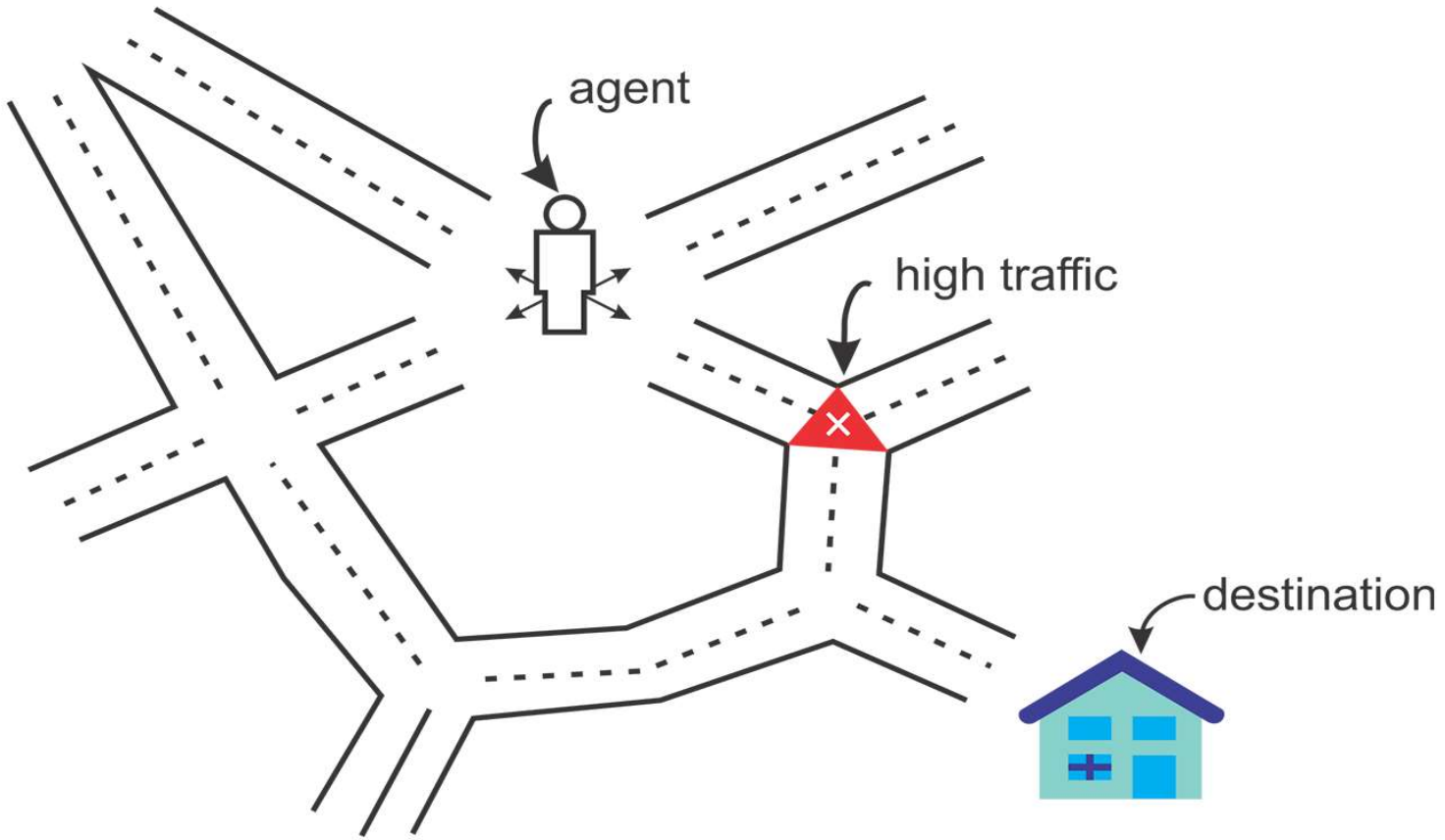


SLIDE 18 // TASK TAXONOMY

Unsupervised



Reinforcement learning



SLIDE 20 // TASK TAXONOMY

Main Categories of ML Tasks



SLIDE 21 // TASK TAXONOMY

Main Categories of ML Tasks



Semi-supervised Learning

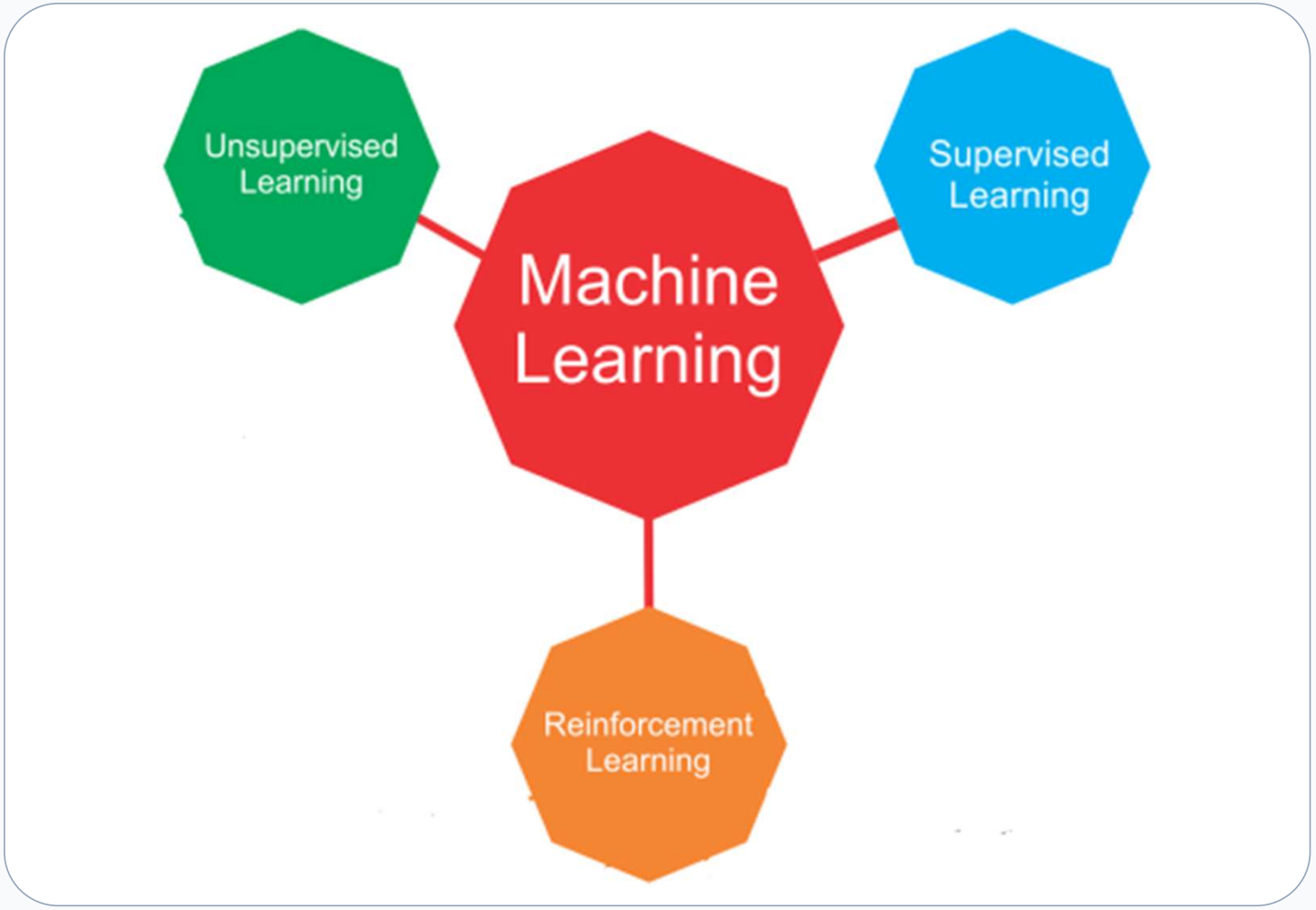
Small labeled set + large unlabeled set

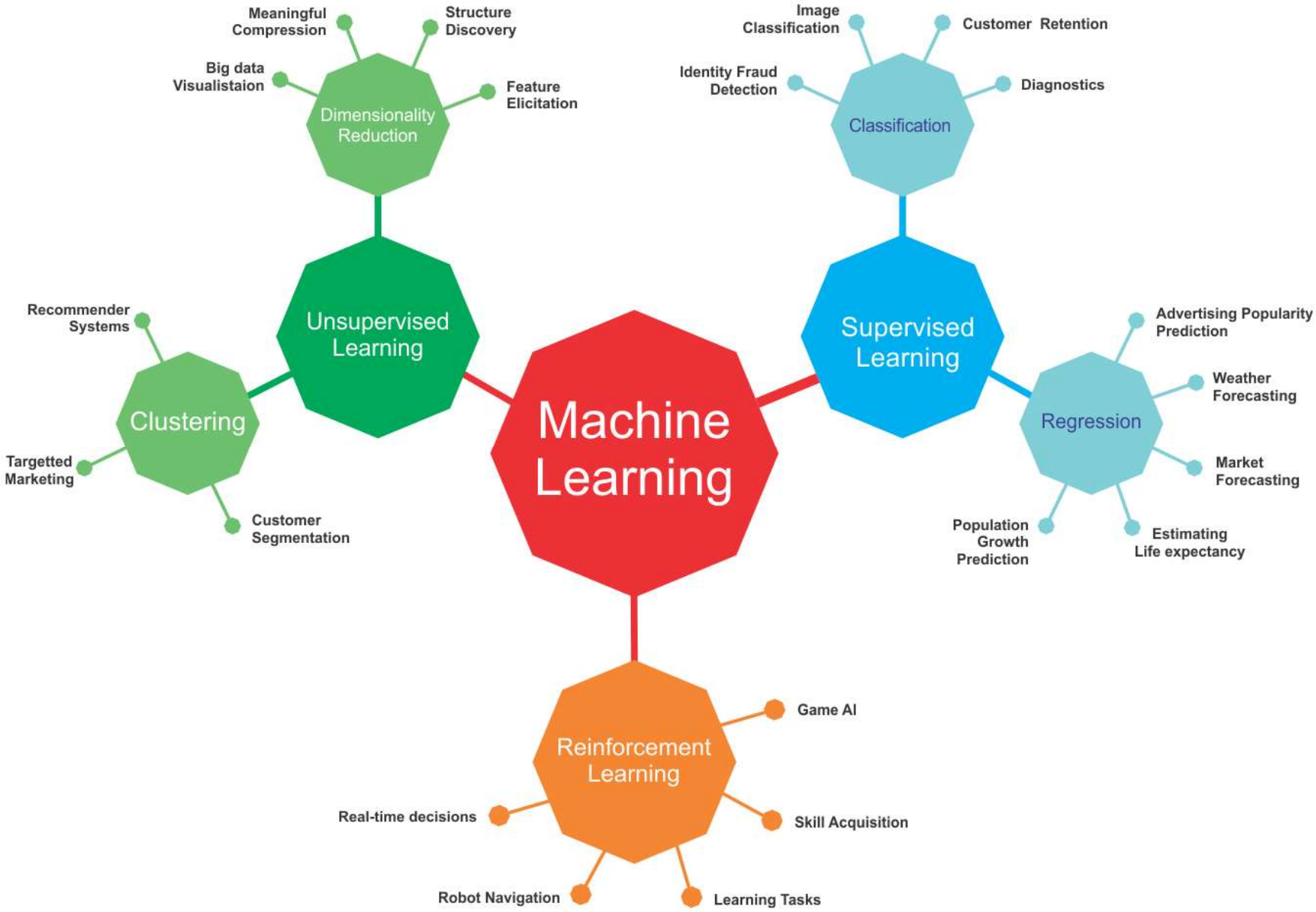
Self-supervised Learning

Supervisory signals extracted from raw data

SLIDE 22 // TASK TAXONOMY

Main Categories of ML Tasks




Machine Learning
Unsupervised Learning
Dimensionality Reduction
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Clustering
Recommender Systems
Targetted Marketing
Customer Segmentation
Supervised Learning
Classification
Image Classification
Customer Retention
Identity Fraud Detection
Diagnostics
Regression
Advertising Popularity Prediction
Weather Forecasting
Market Forecasting
Population Growth Prediction
Estimating Life expectancy
Reinforcement Learning
Game AI
Real-time decisions
Skill Acquisition
Robot Navigation
Learning Tasks

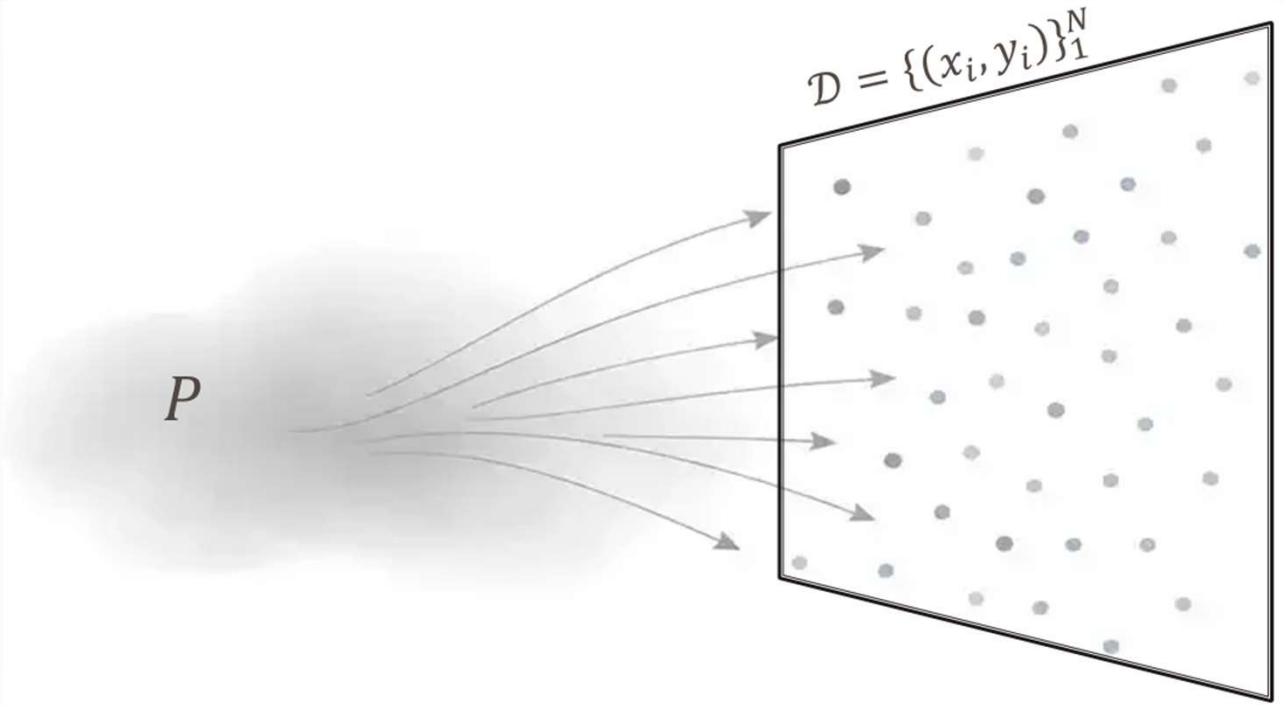
PART 4

Main Components of ML: 3. Model Class



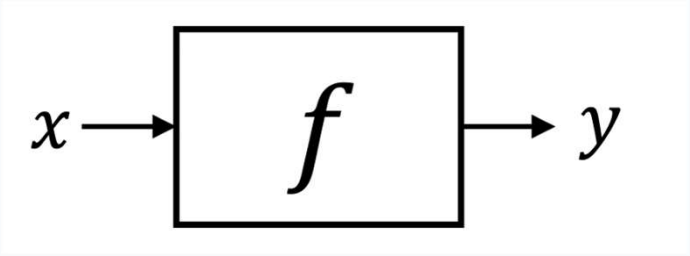
SLIDE 25 // COMPONENT 03

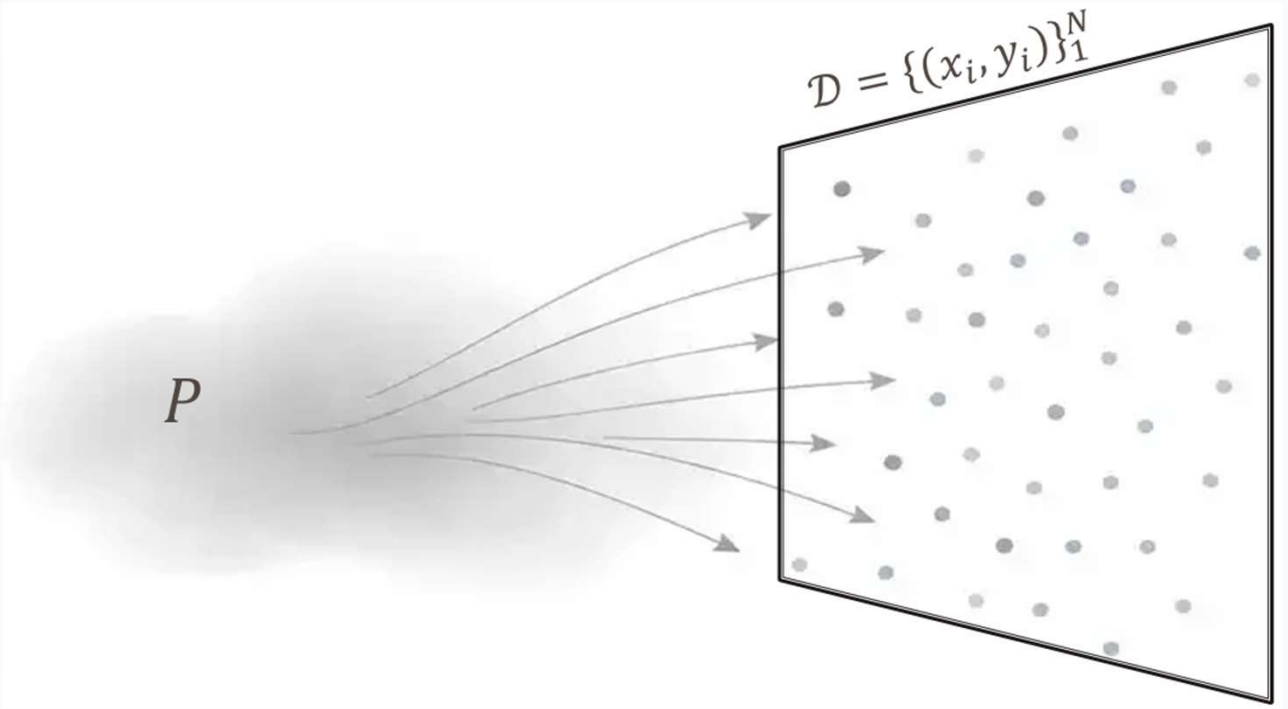
Model Class (Hypothesis Space)



SLIDE 26 // COMPONENT 03

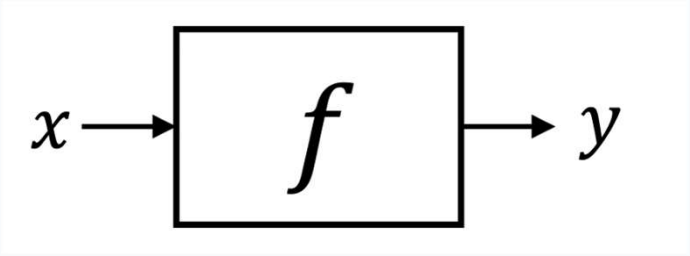
Model Class (Hypothesis Space)





SLIDE 27 // COMPONENT 03

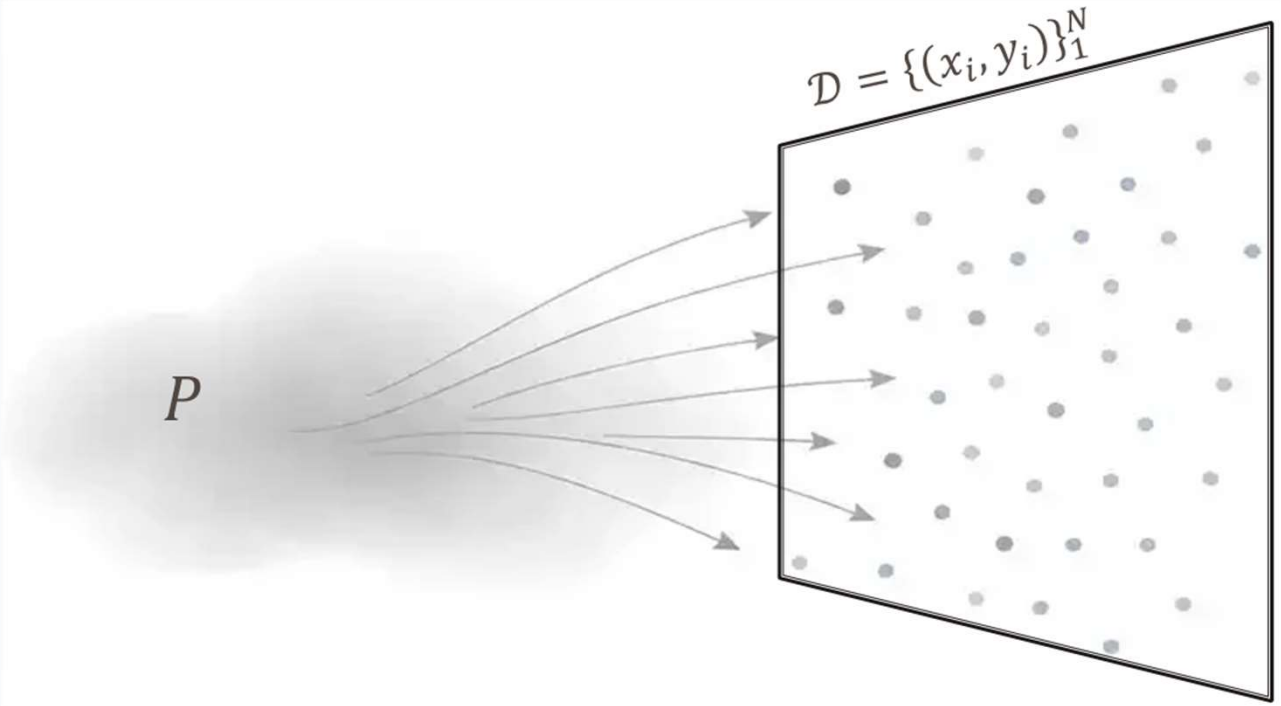
Model Class (Hypothesis Space)





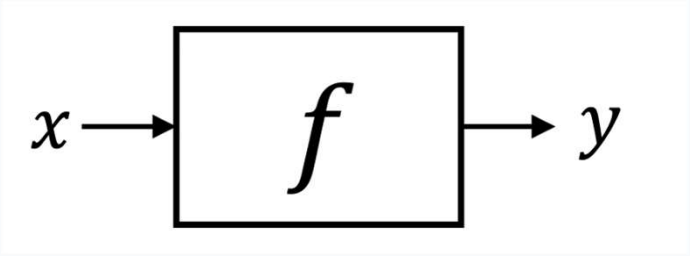
$$\mathcal{D} = \{(x_i, y_i)\}_1^N$$

$$\tilde{f} = f + \epsilon$$



SLIDE 28 // COMPONENT 03

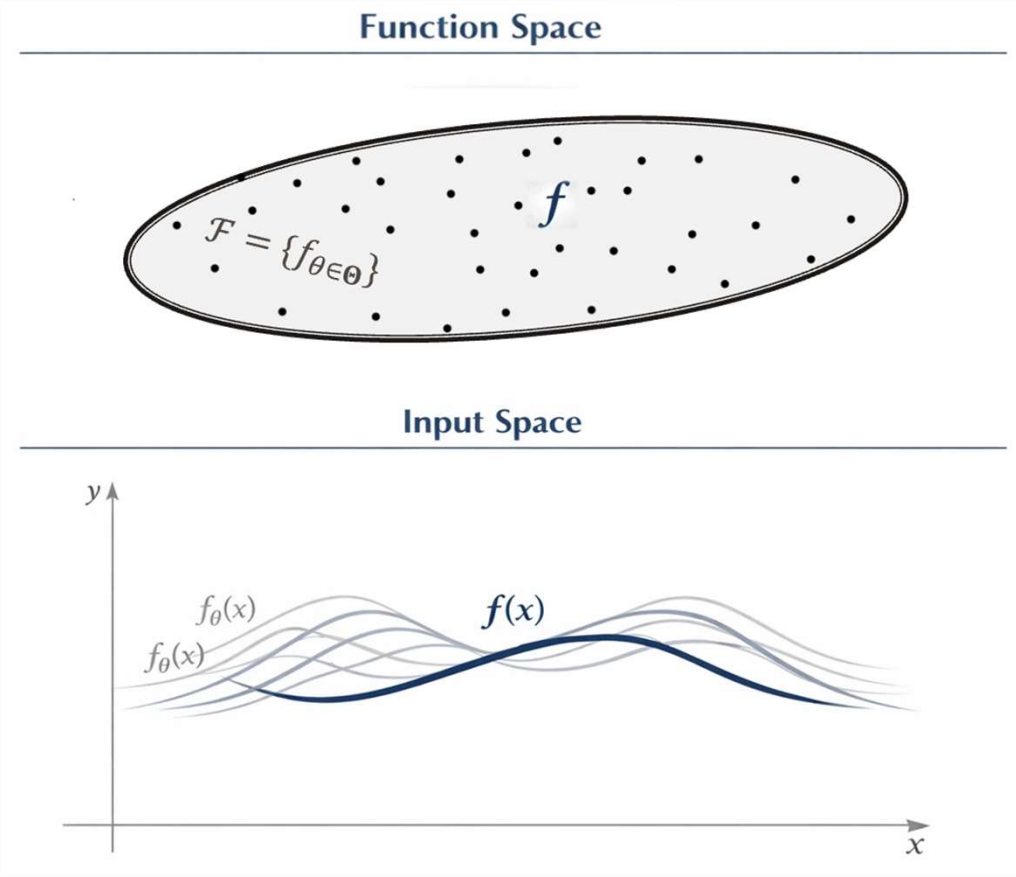
Model Class (Hypothesis Space)





$$\mathcal{D} = \{(x_i, y_i)\}_1^N$$

$$\tilde{f} = f + \epsilon$$



Model Class (Hypothesis Space)

Linear

$$f(x) = \theta_1 x + \theta_0$$

Polynomial

$$f(x) = \theta_0 + \theta_1 x + \ldots + \theta_K x^k$$

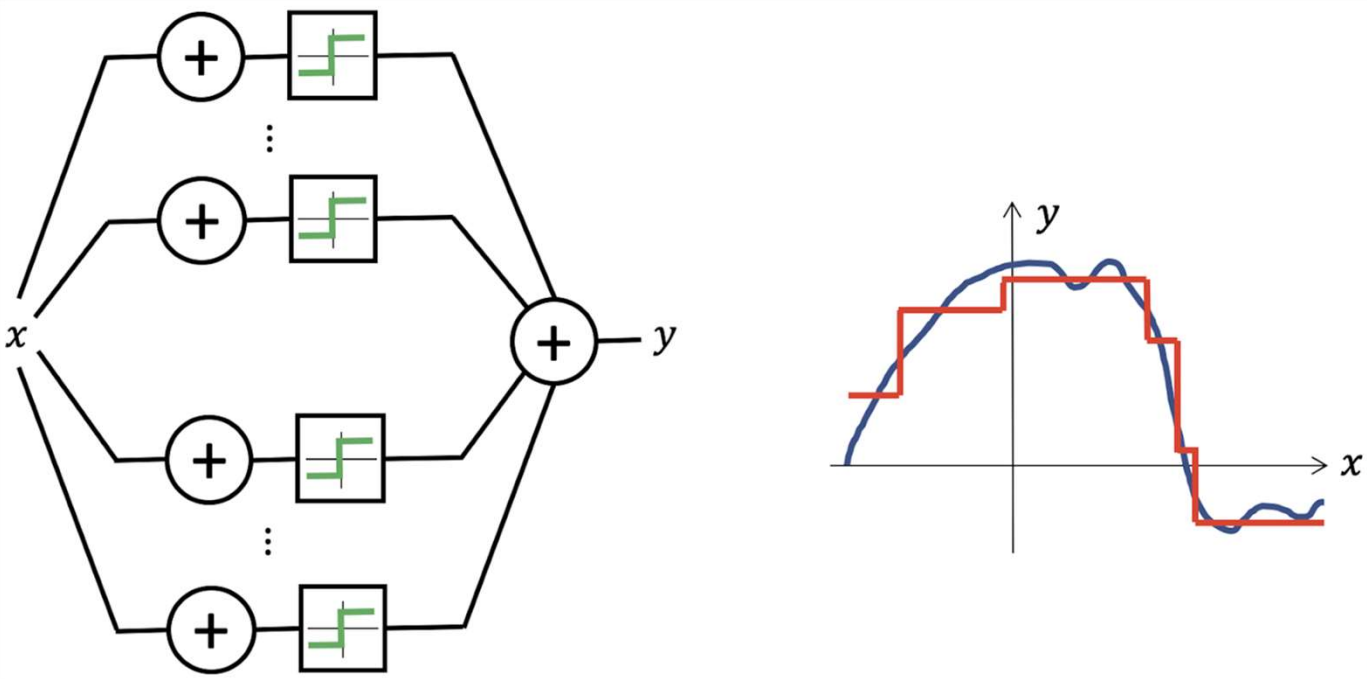
Neural network

$$f(x) = W_2 \sigma(W_1 x + b_1) + b_2$$

SLIDE 30 // COMPONENT 03

Model Class (Hypothesis Space)

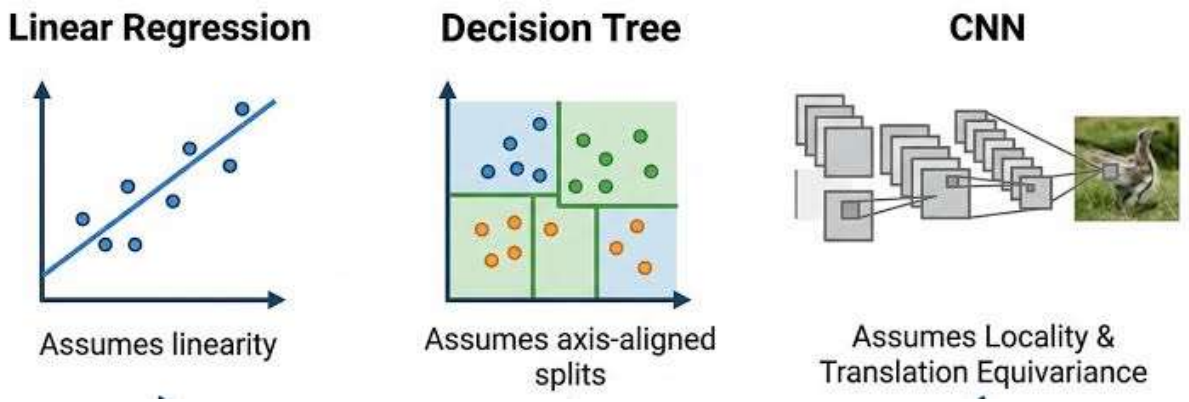
Neural network: MLP



SLIDE 31 // COMPONENT 03

Model Class (Hypothesis Space)

Inductive bias: is the set of assumptions that a machine learning model uses to predict outputs for unseen inputs.



SLIDE 32 // PART 4 – ML MODELS

Kinds of Models

- Linear Models
- Tree-based Models
- Kernel Methods
- Neural Networks
- Probabilistic Models
- Instance-based Models
- Ensemble Methods

SLIDE 33 // PART 4 – ALGORITHMS

Model Examples

Category	Examples
Linear	Linear Regression, Logistic Regression
Tree-based	Decision Tree, Random Forest, XGBoost
Kernel	SVM, Kernel PCA
Neural Networks	MLP, CNN, RNN, Transformer
Probabilistic	Naive Bayes, GMM, HMM
Instance-based	k-NN
Ensemble	Bagging, Boosting

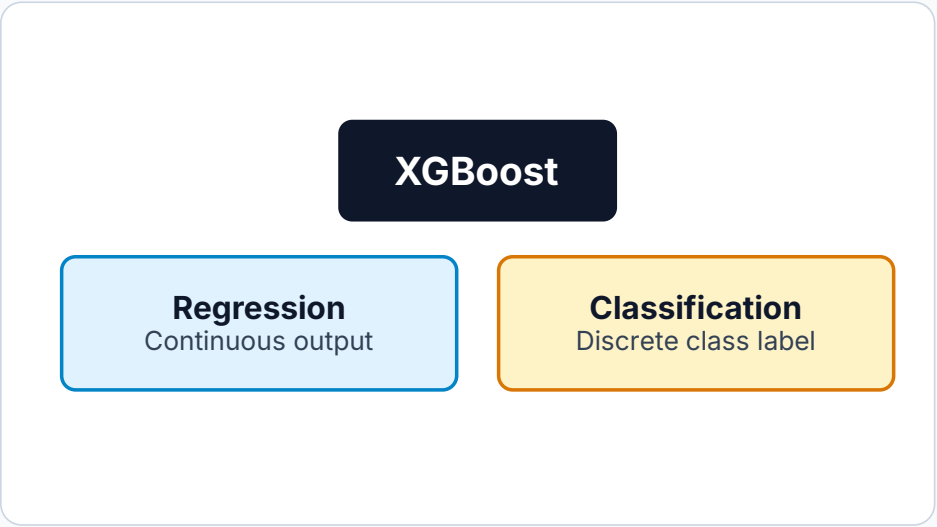
SLIDE 34 // MODEL FLEXIBILITY

One Model, Many Tasks

Dual Capability: Many model families handle both regression and classification

XGBoost Example: Predicts continuous values (regression) or class labels (classification)

Mechanism: Loss function and output layer adapt the model to the task



PART 5

Main Components of ML: 4. Rep. ϕ

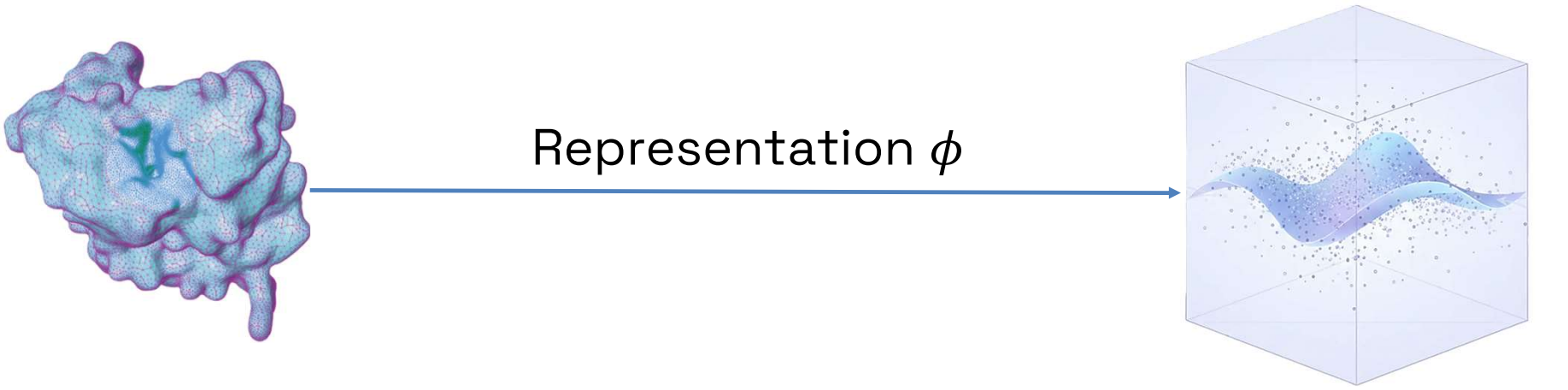


Representation ϕ definition

How do I translate the real world into a mathematical language so that my model has the least difficulty learning patterns?

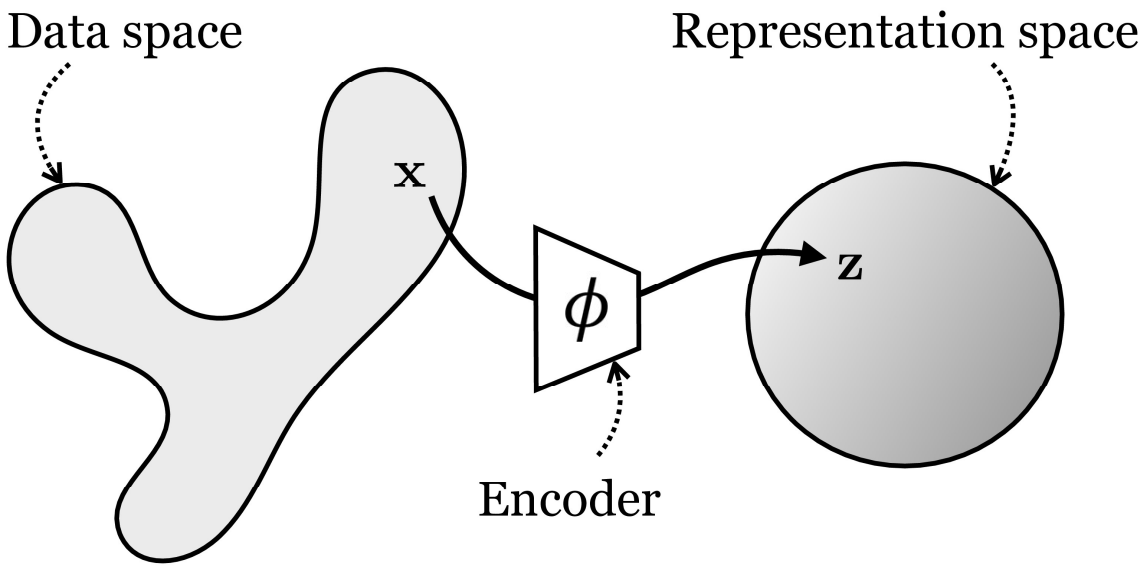
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Representation ϕ definition

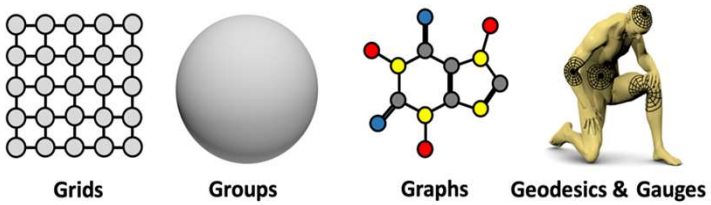
Representation ϕ Transforms raw input x into a dense, structured data $\phi(x)$.



Representation ϕ definition

Representation ϕ Transforms raw input x into a dense, structured data $\phi(x)$.

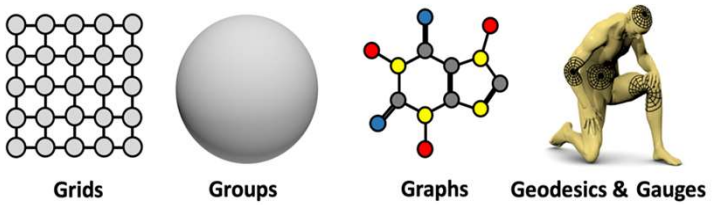
- ϕ Encodes domain invariance such as translation, scale, or token semantics.



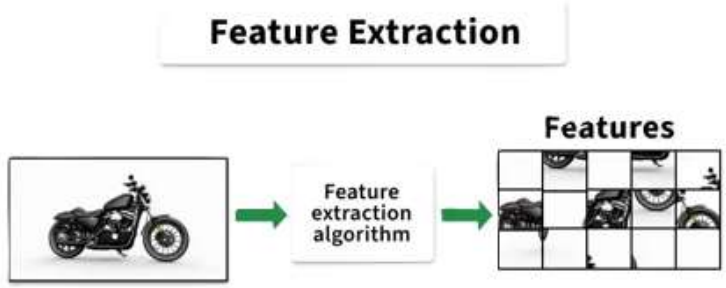
Representation ϕ definition

Representation ϕ Transforms raw input x into a dense, structured data $\phi(x)$.

- ϕ Encodes domain invariance such as translation, scale, or token semantics.



- It can be used as a hand-made, **fixed** transformation or **learned**.



Representation ϕ Fixed vs Learned

Fixed in classic supervised learning

1. One-hot encoding

id	color
1	red
2	blue
3	green
4	blue



id	color_red	color_blue	color_green
1	1	0	0
2	0	1	0
3	0	0	1
4	0	1	0

Representation ϕ Fixed vs Learned

Fixed in classic supervised learning

1. One-hot encoding

id	color
1	red
2	blue
3	green
4	blue



id	color_red	color_blue	color_green
1	1	0	0
2	0	1	0
3	0	0	1
4	0	1	0

Learned in modern representation learning

2. Representation learning in a vanilla CNN poloclub.github.io/cnn-explainer

PART 6

Main Components of ML: 5. Algorithm $\mathcal{A}$

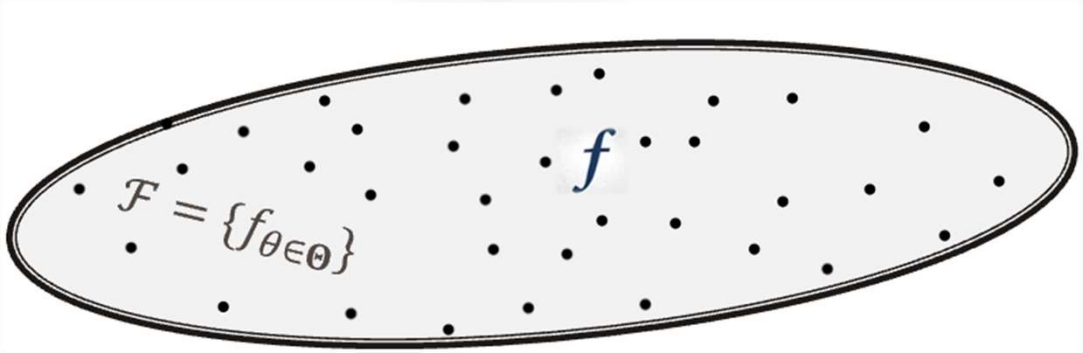


Algorithm $\mathcal{A}$ definition

Algorithm $\mathcal{A}$: The *entire high-level procedure, protocol, or recipe* that takes a dataset $\mathcal{D}$, navigates the hypothesis space $\mathcal{F}$, orchestrates updates/search, handles everything and outputs the final trained model $f^* \in \mathcal{F}$.

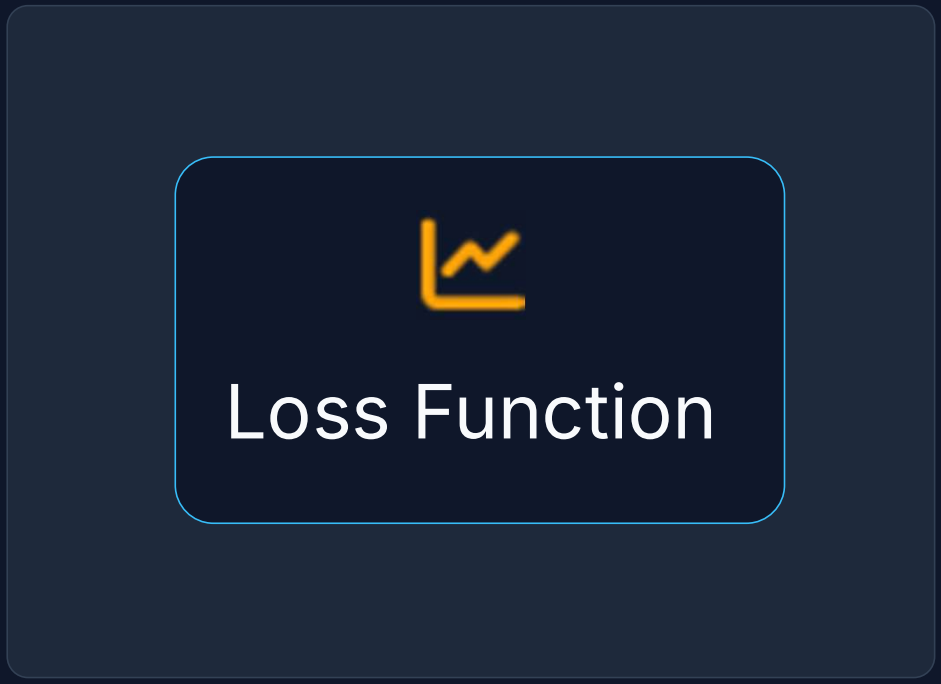
Algorithm $\mathcal{A}$ definition

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PART 7

Main Components of ML: 6. Loss Function



SLIDE 47 // COMPONENT 06

Loss Function

From one prediction to the performance, we care about.

Pointwise Error.
Measures how bad a single prediction is.

$$L\big(y, f_\theta(x)\big)$$

SLIDE 48 // COMPONENT 06

Loss Function

From one prediction to the performance, we care about.

Pointwise Error.
Measures how bad a single prediction is.

$$L\big(y, f_\theta(x)\big)$$

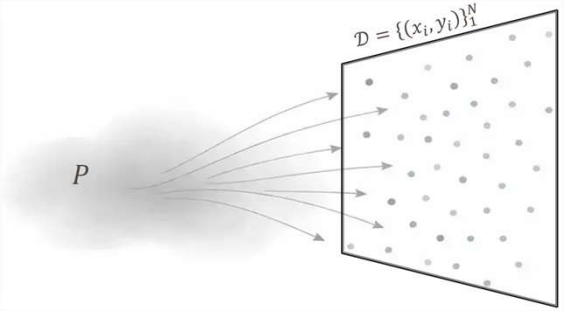
Expected Risk.
Average loss over the true data-generating distribution P—the target we ultimately want to perform well on.

$$\mathcal{R}(f) = \mathrm{E}_{(\mathrm{x},\mathrm{y})\sim P} L\big(y, f(x)\big)$$

SLIDE 49 // COMPONENT 06

Loss Function

From one prediction to the performance, we care about.



Pointwise Error.
Measures how bad a single prediction is.

$$L\big(y, f_\theta(x)\big)$$

Expected Risk.
Average loss over the true data-generating distribution P—the target we ultimately want to perform well on.

$$\mathcal{R}(f) = \mathrm{E}_{(\mathrm{x},\mathrm{y})\sim P} L\big(y, f(x)\big)$$

SLIDE 50 // COMPONENT 06

Loss Function

From one prediction to the performance, we care about.

Pointwise Error.
Measures how bad a single prediction is.

$$L\big(y, f_\theta(x)\big)$$

Expected Risk.
Average loss over the true data-generating distribution P—the target we ultimately want to perform well on.

$$\mathcal{R}(f) = \mathrm{E}_{(\mathrm{x},\mathrm{y})\sim P} L\big(y, f(x)\big)$$

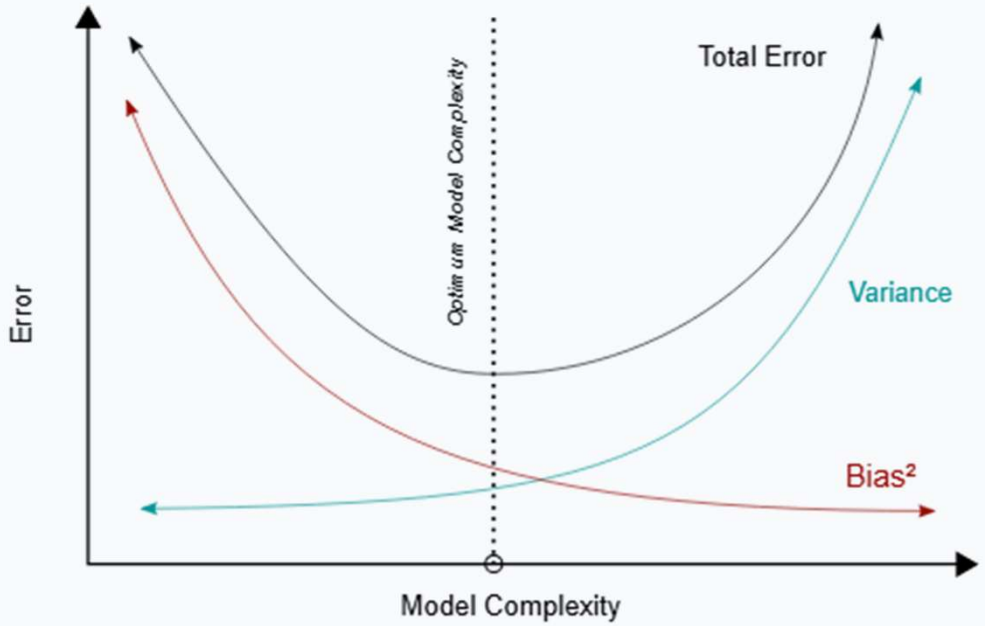
Empirical Risk.
A finite dataset turns the ideal objective into something we can calculate and optimize

$$\mathcal{R}_n(f) = \frac{1}{n}\sum_{1}^{n} L\big(y_i, f_\theta(x_i)\big)$$

SLIDE 51 // COMPONENT 06

bias variance tradeoff

Please google this words and read about.



SLIDE 52 // PART 7 – LOSS FUNCTIONS

Kinds of Loss Functions



Regression Losses



Classification Losses



Ranking Losses



Reconstruction Losses

Kinds of Loss Functions

Regression Losses

Classification Losses

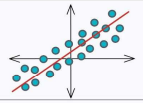
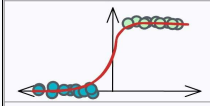
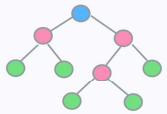
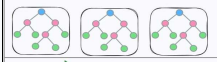
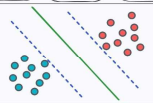
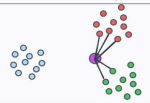
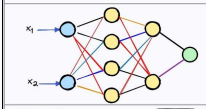
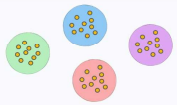
Ranking Losses

Reconstruction Losses

Category	Examples
Regression	MSE, MAE, Huber
Classification	Cross-Entropy, Hinge, Exponential
Ranking	Triplet, Contrastive
Reconstruction	MSE, KL Divergence

Loss functions of 16 Most Popular ML Algorithms

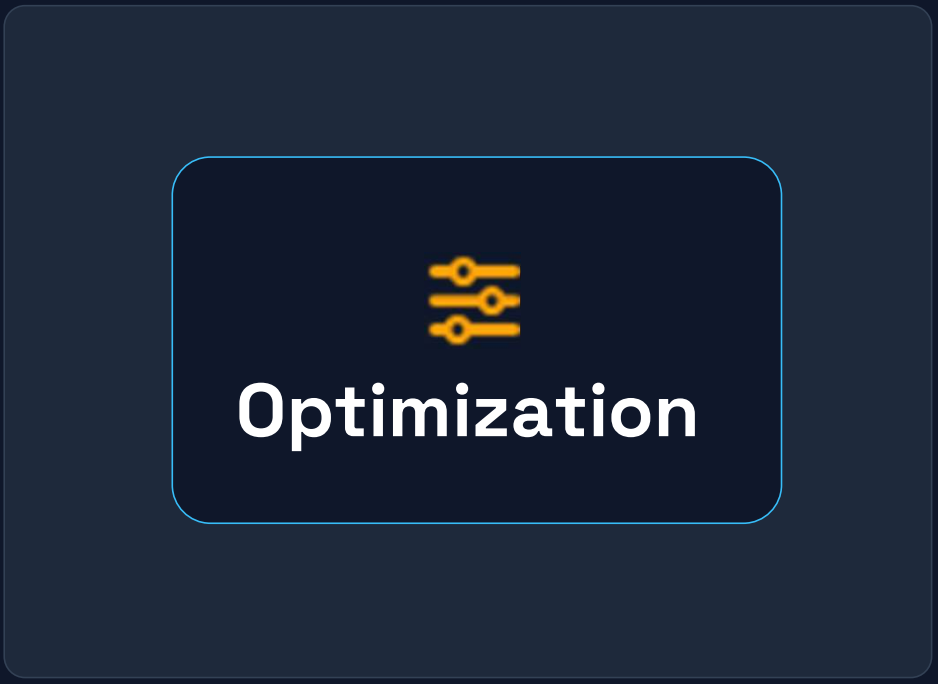
blog.DailyDoseofDS.com

	Linear Regression	Mean Squared Error
	Logistic Regression	Cross-Entropy Loss
	Decision Tree Classifier	Information Gain or Gini impurity
	Decision Tree Regressor	Mean Squared Error
	Random Forest Classifier	Information Gain or Gini impurity
	Random Forest Regressor	Mean Squared Error
	Support Vector Machines (SVMs)	Hinge Loss
	k-Nearest Neighbors	No loss function
$P(B\|A) = \frac{P(B \cap A)}{P(A)}$	Naive Bayes	No loss function
	Neural Networks	Regression: Mean Squared Error Classification: Cross-Entropy Loss
	AdaBoost	Exponential loss
	Gradient Boosting | LightGBM | CatBoost | XGBoost	Regression: Mean Squared Error Classification: Cross-Entropy Loss
	KMeans Clustering	??

*These are typically used loss functions. It does not mean they are used always in these algorithms.

PART 8

Main Components of ML: 7. Optimization



SLIDE 56 // COMPONENT 07

Optimization

Search for parameters minimizing empirical risk:

$$\theta^\star = \underset{\theta \in \Theta}{argmin}\ R_n(f_\theta)$$

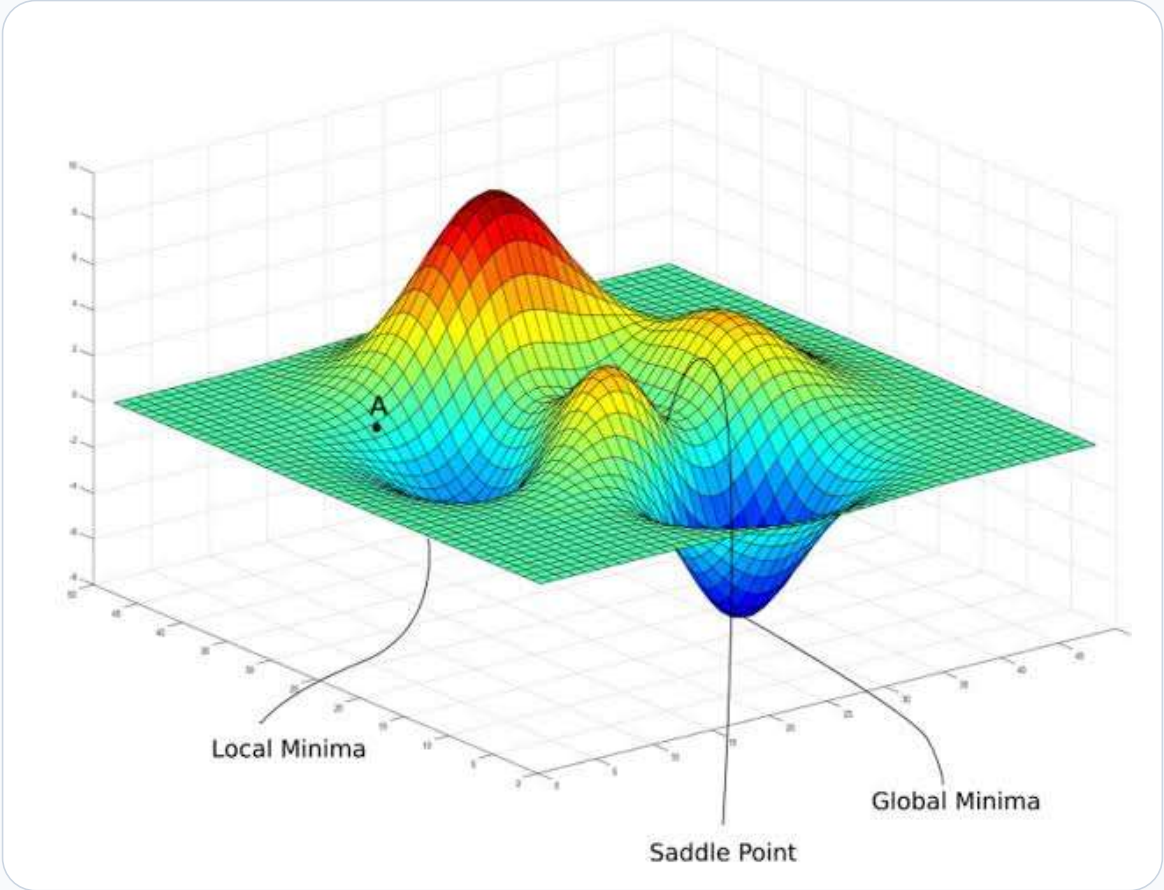
Methods: Gradient Descent, SGD, Adam, L-BFGS



Gradient descent steps converging on a loss surface

Local Minimum Problem

Gradient descent can get stuck in **local minima.**



SLIDE 58 // PART 8 – OPTIMIZATION

Kinds of Optimization



Gradient-based



Second-order



Evolutionary



Convex Optimization

SLIDE 59 // PART 8 – OPTIMIZATION

Kinds of Optimization

Gradient-based

Second-order

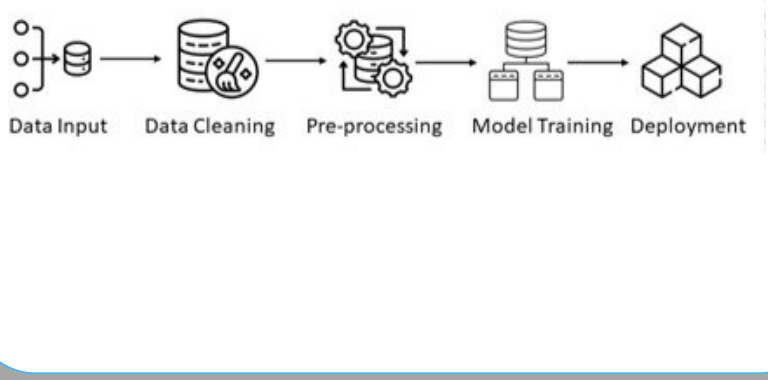
Evolutionary

Convex Optimization

Category	Examples
Gradient-based	Batch GD, SGD, Mini-batch GD, Adam, RMSprop
Second-order	Newton, L-BFGS
Evolutionary	Genetic Algorithms
Convex	Linear Programming, Quadratic Programming

PART 9

Now, how to build a ML pipeline?



SLIDE 61 // PART 9 – SOFTWARE ECOSYSTEM

Python ML Libraries

Data & Arrays

NumPy, pandas

Classic ML

scikit-learn

Visualization

Matplotlib, seaborn

Deep Learning

PyTorch, TensorFlow, Keras

Gradient Boosting

XGBoost, LightGBM

SLIDE 62 // PART 9 – ML PIPELINE

How to Build an ML Pipeline

STEP 01

Problem Definition

Task & success metric

STEP 02

Data Collection

Sources, labeling, ethics

STEP 03

Preprocessing

Cleaning & normalization

STEP 04

Feature Eng.

Domain transformation

SLIDE 63 // PART 9 – ML PIPELINE

How to Build an ML Pipeline

STEP 01
Problem Definition
Task & success metric

STEP 02
Data Collection
Sources, labeling, ethics

STEP 03
Preprocessing
Cleaning & normalization

STEP 04
Feature Eng.
Domain transformation

STEP 05
Model Selection
Hypothesis & biases

STEP 06
Training
Optimize & tune

STEP 07
Evaluation
CV, test set, metrics

STEP 08
Deployment
Serving & monitoring

SLIDE 64 // PART 9 – ML PIPELINE

How to Build an ML Pipeline

STEP 01
Problem Definition
Task & success metric

STEP 02
Data Collection
Sources, labeling, ethics

STEP 03
Preprocessing
Cleaning & normalization

STEP 04
Feature Eng.
Domain transformation

STEP 05
Model Selection
Hypothesis & biases

STEP 06
Training
Optimize & tune

STEP 07
Evaluation
CV, test set, metrics

STEP 08
Deployment
Serving & monitoring



Data → Train → Deploy → Monitor → Retrain

SLIDE 65 // SESSION SYNTHESIS

Summary

Seven Pillars: Dataset, task, model, loss, optimization, representation, algorithm

Rich Taxonomy: Datasets, tasks, models, and losses come in many flavors

Evaluation: Metrics and splits are strictly task-dependent

Execution: The ML pipeline is systematic and iterative

SLIDE 66 // END OF SESSION 03

Session 03 – The ML Landscape